

Day-7

17-04-2026

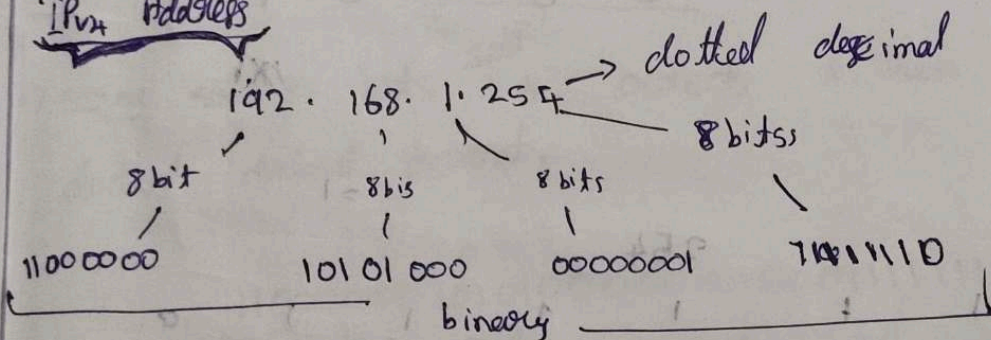
IPV4 ADDRESSING

Part-I

OSI-model - Network Layer

- ⇒ Provides connectivity between end hosts on different networks (outside of the LAN)
- ⇒ provide Logical Addressing (IP Addresses)
- ⇒ Provides path selection between source & destination
- ⇒ Routers operate at Layer 3

IPv4 Address



Decimal & Hexadecimal

Decimal
(base 10)

3 2 9 4

3×1000 2×100 9×10 4×1

Hexadecimal
(base 16)

C D E F

$C \times 256$ $D \times 16$ $E \times 1$ ($E=14$)

($C=12$) ($D=13$) $F=15$

3072 + 208 + 14 = 3294

Binary

Change 2)

1 1 0 0 0 0 0 0
1x128 1x64 0x32 0x16 0x8 0x4 0x2 0x1

$$128 + 64 = 192$$

168

1 0 1 0 0 0 0 0
1x128 0x64 1x32 0x16 0x8 0x4 0x2 0x1

$$128 + 32 + 8 = 168$$

0 0 0 0 0 0 0 1

1 1 1 1 1 1 1 0
1x128 1x64 1x32 1x16 1x8 1x4 1x2 0x1

$$128 + 64 + 32 + 16 + 8 + 4 + 2 = 254$$

Decimal to Binary

221

128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1
221	93	28	28	12	4	0	0
-128	-64	-32	-16	-8	-4	-2	-1
=93	=28	=0	=12	=4	=0	=0	=0
1	1	0	1	1	0	0	0

127

128	64	32	16	8	4	2	1
↓	↓	↓	↓	↓	↓	↓	↓
127	127	63	31	15	7	3	1
-128	-64	-32	-16	-8	-4	-2	-1
=0	=63	=31	=15	=7	=3	=1	=0
0	1	1	1	1	1	1	0

00000000 = 0
11111111 = 255

IPv4 Address

⇒ series of 32 bits split into 4 octets & then written in dotted decimal format.

11000000 10101000 00000000 11111110
↓ ↓ ↓ ↓
11000000 10101000 00000001 11111110
↓ ↓ ↓ ↓
192 168 1 254 /24

24 means 24 bits. First 24 bits of this IP address represent the network portion of the address & remaining 8 represent end host

10011010010011100110111100100000

$$10011010 = 128 + 16 + 8 + 2 = 154$$

$$01001110 = 64 + 8 + 4 + 2 = 78$$

$$01101111 = 64 + 32 + 8 + 4 + 2 + 1 = 111$$

$$00100000 = 32$$

$$154 \cdot 78 \cdot 111 \cdot 32$$

NW Address

→ Host portion of the address is all 0's = NA

→ NA cannot be assigned to be a host.

Broadcast Address

→ Host portion of the address is all 1's = Broadcast Address

→ The BC Address cannot be assigned to a host

IPv4 Address Classes

Class	First octet	First octet numeric range	Prefix length
A	0xxxxxx	0 - 127	8
B	10xxxxxx	128 - 191	16
C	110xxxxx	192 - 223	24
D	1110xxxx	224 - 239 - multicast	
E	1111xxxx	240 - 255 - Experimental	

Loop back Addresses

• Address Range 127.0.0.0 - 127.255.255.255

• used to test the 'network stack' (think OSI, TCP/IP model) on the local device.

Day-8

18-04-26

IPv4 ADDRESSING

PART - II

Maximum Hosts per network

$$MHPN = 2^n - 2 \quad (n = \text{number of host bits})$$

Class A

$$192.168.1.0 / 24 \longrightarrow 192.168.1.255 / 24$$

↓

$$\text{Host portion} = 8 \text{ bits} = 2^8 = 256$$

↓

Host portion all 0
= NWA

$$MHPN = 2^8 - 2 = 254$$

Host portion all 1s
= broadcast address

Class B

$$172.16.0.0 / 16 \longrightarrow 172.16.255.255 / 16$$

$$\text{Host portion} = 16 \text{ bits} = 2^{16} = 65,536$$

$$MHPN = 2^{16} - 2 = 65,534$$

First/Last usable Address:

$$192.168.1.0 / 24 \longrightarrow 192.168.1.255 / 24$$

00000000

↓

00000001

192.168.1.1 / 24

First usable Address

11111111

↓

11111110

192.168.1.254 / 24

Last usable address